YANXAERO: Dual-Mode Aerospace Engine Combining Linear Rotating Detonation and Nonlinear Electromagnetic Pulse Stepped Transient Propulsion

琰擎-线性旋转爆震与非线性电磁脉冲步进瞬跃推进的双模空天发动机

Based on CDEQ V9.0 Spatiotemporal Energy Cooperative Control Theory

基于 CDEQ V9.0 时空能量协同调控理

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Abstract

As a core technical pathway for pressure-gain combustion, the rotating detonation engine has long been constrained by three fundamental dilemmas in its engineering translation: incompatibility between stability and propulsion, contradiction between high-power output and system robustness, and the gap between short-term stability and long-term reliability. Traditional control theory separates stability analysis from propulsion design, failing to resolve multi-objective, strongly nonlinear, and full-cycle coupling problems within a unified mathematical framework.

This paper presents the YANXAERO dual-mode aerospace engine, built on the theoretical core of CDEQ V9.0, the stable kernel of the V5.0 Harpoon Equation, and the V7.0 Functional Optimal Navigation as the decision-making brain, integrating Hua Luogeng's 0.618 optimization method into an eight-stage optimal control framework. The system achieves dual-mode fusion of linear rotating detonation propulsion (atmospheric phase) and nonlinear electromagnetic pulse stepped transient propulsion (space phase). It adopts the Penta-Cyclic-Clockwise-Topological-Theorem (PCTT-CW) as the dynamic evolution backbone, the Central-Holding-Nonary-Constraint-Execution-Theorem (CENHE-LEX) as the global dynamic constraint, and the Global-Convergence-Validation-Theorem (GCVT-10) ten-step convergence mechanism as the safety interlock guarantee. It strictly complies with the PENTAXCORE four-level unidirectional non-preemptive architecture and a 167 ms unified time lattice, realizing unified control in the temporal, energy, and spatial domains. This paper also proves the global Lyapunov stability of the PCTT-CW topology, mathematically verifying the causal structure and convergence properties of the system.

For the first time, this work achieves strict isomorphism between the PCTT-CW topology and the physical structure of the YANXAERO engine, mapping the five-element topology directly into a five-stage physical structure: intake, compression, ignition, expansion, and exhaust. A mechanism of nearest-neighbor excitation and next-neighbor damping is established for ordered energy transfer and dynamic balancing. Based on Hua Luogeng's optimization method, the eight-stage optimal thrust coefficient search is completed with minimal experimental effort. For performance degradation caused by long-term engine operation, three full-life thrust parameter sets (new, mid-life, aged) are preconfigured to achieve health-adaptive optimal control. The system employs a 20° V-shaped dual-oscillator coupling topology, a rounded engineering value of the golden-section angle, enabling synergistic coupling between forward energy conversion and phase synchronization monitoring.

The CDEQ V9.0 framework strictly conforms to IEC 61508 SIL-4 functional safety, DO-254 avionics hardware assurance, and IEEE 1012 software verification standards. All control logic is verifiable, synthesizable, certifiable, and engineering-ready. This theoretical system elevates YANXAERO from passive stability to active decision-making, from trial-and-error to theoretical optimality, and from fixed parameters to full-life adaptation, providing a complete, rigorous, and engineerable theoretical foundation for aerospace propulsion.

Keywords:
YANXAERO engine; CDEQ V9.0; linear rotating detonation; nonlinear electromagnetic pulse; stepped transient; dual-mode propulsion; eight-stage control; PCTT-CW; CENHE-LEX; GCVT-10; full-life adaptive control

1. Problem Statement

1.1 Engineering State of the Art in Rotating Detonation Engines

Rotating detonation engines achieve an approximate constant-volume combustion cycle based on the extreme thermodynamics of detonation combustion, with typical detonation wave speeds exceeding 1500 m/s and theoretical thermal efficiency up to 49%, significantly higher than the 27% of conventional isobaric combustion. After decades of laboratory research, basic detonation wave establishment and short-duration operation have been demonstrated. However, in the transition from laboratory prototypes to engineering systems, persistent challenges remain: insufficient initiation reliability, narrow stable propagation boundaries, unsteady oscillatory instability, and thermal efficiency losses due to parasitic combustion.

Current research focuses heavily on configurations and combustion mechanisms, while control system development lags severely. As a result, engines can only operate briefly under narrow, fixed conditions and cannot adapt to variable thrust, variable operating points, long-duration, or full-life engineering requirements.

1.2 Three Core Control Challenges

Challenge 1: Stability and actuation cannot be reconciled.
The engine can maintain stable detonation wave structure at fixed equivalence ratios and mass fluxes, but dynamic commands such as thrust modulation, mode switching, or phase shifting immediately induce detonation wave drift, pressure oscillation overruns, thermoacoustic instability, or even quenching.

Challenge 2: High-power output conflicts with structural stability.
Under high-thrust conditions, the probability of detonation wave phase runaway rises sharply. Pressure pulses inside the combustor can exceed tens of atmospheres, and thermal stress concentration accelerates wall ablation. Thrust escalation often coincides with system collapse.

Challenge 3: Short-term stability does not guarantee long-term reliability.
Some laboratory prototypes achieve approximate stability briefly, but without long-term convergence validation. Unpredicted quenching, wave structure breakdown, or structural ablation often occur after minutes of continuous operation.

1.3 Nature of the Engineering Problem

The control problem of rotating detonation engines is a typical multi-stage, multi-objective, full-life, strongly nonlinear coupled optimization problem. The engine operates across eight distinct phases: start, warm-up, initiation, climb, peak, hold, decay, and reset. Over its service life, it degrades through new, mid-life, and aged conditions. Traditional control uses fixed parameters for all operating conditions and lifecycles, inevitably leading to overshoot in new engines, adequate performance in mid-life, and insufficient output in aged units.

1.4 Core Scientific Question

Can a unified theoretical framework be constructed to transform rotating detonation engines from passive stability to active decision-making, from experimental trial-and-error to theoretical optimality, from fixed parameters to full-life adaptation,

and from separate structure and control to a unified paradigm where topology defines structure and structure defines control?

2. Analysis: Five Fundamental Physical Pillars

The CDEQ V9.0 system is rigorously anchored to five classical physical laws:

- Einstein's mass–energy equivalence $E=mc^2$: Defines the upper bound of energy conversion in detonation combustion, eliminating designs exceeding physical limits.
- Maxwell's equations: Provide the mathematical basis for field evolution and momentum transport in electromagnetic propulsion.
- Noether's theorem: Establishes the correspondence between symmetry and conservation laws, ensuring structural invariance under degradation and variable conditions.
- Newton's three laws: Form the foundation for thrust output and motion response.
- Lyapunov stability theory: Provides the mathematical criterion for global stability, underpinning the GCVT-10 convergence mechanism.

3. Three Cornerstones: Eternal Foundations of the CDEQ Framework

3.1 Penta-Cyclic-Clockwise-Topological-Theorem (PCTT-CW)

The PCTT-CW is the minimal, complete, globally optimal, and uniquely stable cyclic topology for system dynamics, governed by nearest-neighbor excitation and next-neighbor damping. **It is a clockwise, unidirectional, irreversible topology** following the cycle 12→3→6→9→12. It is the only cyclic structure that simultaneously enables forward energy transfer, reverse overload constraint, and global dynamic balance. Topologies with 3, 4, or 6 nodes cannot achieve global convergence.

Five-element evolution formula:
$A_i(t) = S\cdot P_{i-1}(t-1) - C\cdot P_{i-2}(t-2) + W(t)\cdot \xi_i(t)$
$P_i(t) = \mathrm{clip}(P_i(t-1)+A_i(t), 0.0, 1.0)$

where:

- $S=0.618$: nearest-neighbor excitation coefficient
- $C=0.382$: next-neighbor damping coefficient
- $A_i(t)$: state increment
- $P_i(t)$: normalized state
- $W(t)\cdot \xi_i(t)$: operational disturbance
- clip(·): amplitude limiting function

Nearest-neighbor excitation: Adjacent elements mutually reinforce to sustain evolutionary activity.
Next-neighbor damping: Elements separated by one node mutually suppress to prevent state overflow and unbounded divergence.

3.2 Central-Holding-Nonary-Constraint-Execution-Theorem (CENHE-LEX)

CENHE-LEX is the highest-priority dynamic constraint unit, derived from the self-righting principle: the central anchor is fixed at 5, and any deviation induces a restoring force proportional to the offset, ensuring the system always returns to

equilibrium.
Base matrix:
$L = [[4,9,2],[3,5,7],[8,1,6]]$

Core constraints:

- Central value anchored at 5
- Row, column, and diagonal sums dynamically approach 15

Central anchor correction:
$\nabla_center = 0.3 \cdot (5.0 - E_22_current)$

Global conservation correction:
$\nabla_ij = 0.1 \cdot ((15 - row_sum_i) + (15 - col_sum_j))$

Functions:

1. Eliminate drift via central anchoring
2. Preserve global structural conservation
3. Suppress and recover from overload states
4. Block non-compliant operations

3.3 Global-Convergence-Validation-Theorem (GCVT-10)

GCVT-10 is a mandatory 10-step consecutive convergence verification mechanism that eliminates pseudo-stability, false activation, and short-duration steady states. It complies with DO-178C, IEC 61508 SIL-4, and IEEE 1012.

Cost function:
$\Phi = \Sigma[\ \|D(t) - 0.8 \cdot Dev(t)\|^2 + 0.2 \cdot \|P(t) - P(t-1)\|^2\ ]$

Convergence criteria (must hold for 10 consecutive steps):

- Central state deviation: $|E_11 - 5.0| < \varepsilon$
- Structural balance deviation: $\max(|row_sum - 15|, |col_sum - 15|) < \varepsilon$
- Cost function: $\Phi < \Gamma$

3.4 Lyapunov Stability Proof of the PCTT-CW Causal Ring

3.4.1 System Definition

Five discrete states: P_0, P_1, P_2, P_3, P_4. The system forms a strictly causal, closed clockwise topology, with state updates depending only on historical information and no reliance on future, ungenerated data.

3.4.2 Structural Law: Explicit Closure via Modulo-5 Constraint

Subscripts follow modulo-5 cyclic arithmetic:

- $P_((i-1) \bmod 5)$: direct predecessor (parent node)

- $P_{((i-2) \bmod 5)}$: indirect predecessor (grandparent node)

Explicit closure:
$P_{-1} \equiv P_4$, $P_{-2} \equiv P_3$, $P_5 \equiv P_0$

State update:
$P_i(t+1) = \mathrm{clip}(P_i(t) + S \cdot P_{((i-1) \bmod 5)}(t) - C \cdot P_{((i-2) \bmod 5)}(t), 0.0, 1.0)$

3.4.3 Proof of Closed Topology (Not a Linear Chain)

Nearest-neighbor synchronization energy:
$V_1 = 0.5((P_1-P_0)^2 + (P_2-P_1)^2 + (P_3-P_2)^2 + (P_4-P_3)^2 + (P_0-P_4)^2)$

The term $(P_0-P_4)^2$ explicitly closes the ring and rules out an open linear structure.

3.4.4 Global Lyapunov Function

$V = V_1 + V_2 + V_3$

$V_1 = 0.5 \cdot \Sigma(P_i - P_{((i-1) \bmod 5)})^2$
$V_2 = \Sigma(P_i - S \cdot P_{((i-1) \bmod 5)} + C \cdot P_{((i-2) \bmod 5)})^2$
$V_3 = 0.5(E_{22}-5)^2 + 0.1 \cdot \Sigma(\mathrm{row_sum}-15)^2 + 0.1 \cdot \Sigma(\mathrm{col_sum}-15)^2$

3.4.5 Stability Verification

The system satisfies global asymptotic stability:

1. $V \geq 0$
2. $dV/dt \leq 0$ under excitation–damping dynamics
3. $V=0$ and $dV/dt=0$ only at the global equilibrium

Eigenvalue verification: All eigenvalues lie inside the unit circle. Reversing excitation and damping causes eigenvalues to exit the unit circle, leading to divergence. The sign convention is uniquely determined by stability.

3.4.6 Final Conclusion

1. Modulo-5 + closed V_1 = rigid five-element ring, not open chain.
2. Nearest-neighbor excitation, next-neighbor damping = unique stable solution.
3. Strict causality, dependence only on historical data = physically realizable and engineerable.
4. $V \geq 0$, $dV/dt \leq 0$ = global Lyapunov stability.

4. Physical Structure and Isomorphism

4.1 Five-Stage Physical Isomorphism

- Element 0: Intake
- Element 1: Compression
- Element 2: Ignition

- Element 3: Expansion
- Element 4: Exhaust

4.2 20° V-Shaped Dual-Oscillator Coupling

$C_V \approx [[0.9397, 0.3420],[0.3420, 0.9397]]$

where cos20° corresponds to nearest-neighbor excitation, sin20° to next-neighbor damping.

4.3 Nearest-Neighbor Excitation: Ordered Energy Transfer

Intake supports compression; compression supports ignition; ignition supports detonation; detonation supports exhaust; exhaust supports intake.

4.4 Next-Neighbor Damping: Dynamic Restraint Mechanism

Excessive intake suppresses ignition; over-compression interferes with detonation; over-ignition hinders exhaust; detonation backpressure disturbs intake; poor exhaust weakens compression.

5. CDEQ Version List and Core Equations

5.1 CDEQ Version List

- V2.0: Stability and basic braking logic
- V3.5: Critical transition and throttle switching
- V5.0: Harpoon equation – stable-drive integration kernel
- V6.0: TANXLON 20° V-shaped dual-oscillator topology, QTXS stepped transient theory
- V7.0: Functional optimal navigation equation
- V8.0: Electromagnetic pulse propulsion main equation
- V9.0: Three-domain unified master equation

5.2 V5.0 Harpoon Equation

$\Psi' = \mathrm{clip}(\varphi\cdot\Psi_1 - \bar{\varphi}\cdot\Psi_2 + \omega\cdot\xi) + (\Psi\geq\Psi\pi \wedge \Pi\mathrm{self}\geq\Pi\mathrm{ext})\cdot\nabla C$

5.3 V7.0 Functional Optimal Navigation Equation

$J = \int(t_0\rightarrow t_f)[\alpha\cdot R(t)+\beta]dt + \gamma\cdot\|X(t_f)-X_target\|^2$

5.4 V8.0 Electromagnetic Pulse Propulsion

$F = 2\cdot\eta\cdot A\cdot E\cdot H / c$

5.5 CDEQ V9.0 Three-Domain Unified Master Equation

$\Psi' = \mathrm{clip}(\varphi\cdot\Psi_1 - \bar{\varphi}\cdot\Psi_2 + \omega\cdot\xi) + (\Psi\geq\Psi\pi \wedge GCVT_{10})\cdot[\nabla C + \kappa\cdot\Delta x\cdot\cos 20°]$

6. Dual-Mode Switching Logic

- Atmospheric mode: rotating detonation, 0–40 km
- Space mode: electromagnetic pulse stepped transient propulsion, above 40 km

Switching is performed only after GCVT-10 convergence and at 167 ms time-lattice boundaries.

7. PENTAXCORE Four-Level Non-Preemptive Architecture

- L4 Mission Decision
- L3 Strategic Planning
- L2 Tactical Scheduling
- L1 Real-Time Execution

Hierarchy: L4 > L3 > L2 > L1. Unidirectional command flow, non-preemptive execution, fault-domain isolation, hardware-level fallback.

Base time lattice: 167 ms.

8. Eight-Stage Cycle & 0.618 Optimization

Eight standardized stages: Init, Warm-up, Stroke, Climb, Peak, Hold, Decay, Reset.
Hua Luogeng's 0.618 method is used to find the global optimal thrust coefficient with minimal testing.

9. Full-Life Adaptive Parameter Sets

Three health-adaptive parameter sets are predefined:

- New Engine Parameter Set
- Mid-Life Engine Parameter Set
- Aged Engine Parameter Set

Health status is evaluated online, and the system switches sets adaptively to maintain optimal performance throughout the service life.

10. Hardware Implementation Principles

- Core: No branching, no floating-point errors, hardware fault isolation, hardware-level fallback.
- Time lattice: 167 ms hardware clock, deterministic and jitter-free.
- Redundancy: Multi-core heterogeneous voting at each level.

11. Conclusion

This paper establishes a complete control paradigm for rotating detonation engines based on CDEQ V9.0 and YANXAERO. The nearest-neighbor excitation and next-neighbor damping mechanism embedded in the PCTT-CW topology fundamentally resolves instabilities such as wave drift, thermoacoustic coupling, and surge. The Lyapunov stability proof provides a rigorous mathematical foundation. The full-life adaptive strategy and deterministic hierarchical architecture enable long-duration, high-reliability operation. This framework offers a practical, structurally simple, and controllable pathway for next-generation aerospace propulsion.

12. Intellectual Property Statement

The YANXAERO engine paradigm, CDEQ V9.0 framework, **Penta-Cyclic-Clockwise-Topological-Theorem (PCTT-CW), Central-Holding-Nonary-Constraint-Execution-Theorem (CENHE-LEX), Global-Convergence-Validation-Theorem (GCVT)**, V5.0–V9.0 equations, 20° V-shaped dual-oscillator topology, eight-stage control law, full-life adaptive method, 167 ms time lattice, and PENTAXCORE architecture are the original intellectual property of Wenjia Jiang. Any use, citation, implementation, or extension requires full attribution.

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The Dao of CDEQ

Heart as Compass, Sword of Readiness.
Seek no trouble, fear no trouble.
Sword in sheath, hand on hilt.
Heart as anchor, mouth without noise.
Ignore us now.
The gate was never open.